

FABRICATION SPECIFICATION

Project: USB3 Fiber Link

GENERAL

Layers: 4 (TOP / GND / PWR / BOTTOM)
Material: FR4, Standard Tg (150–160C)
Finished Thickness: 62 mil (1.575mm) +/- 10%
Surface Finish: ENIG (1u")
Solder Mask: Green, both sides
Silkscreen: White, both sides
IPC Class: 2

STACKUP

F.Cu: 1 oz (0.035mm) – Signal
Prepreg: 0.21mm (7628, Er=4.5)
In1.Cu: 0.5 oz (0.0175mm) – GND Plane
Core: 1.03mm (Er=4.5)
In2.Cu: 0.5 oz (0.0175mm) – PWR Plane
Prepreg: 0.21mm (7628, Er=4.5)
B.Cu: 1 oz (0.035mm) – Signal

CONTROLLED IMPEDANCE

USB3 SuperSpeed: 100 ohm diff, 50 ohm SE
SFP+ CML: 100 ohm diff, 50 ohm SE
USB2 High-Speed: 90 ohm diff, 45 ohm SE
Tolerance: +/-10% (+/-5 ohm if <= 50 ohm)
Verify trace width with fab stackup calculator

MINIMUM FEATURES (JLPCB 4-layer 1oz)

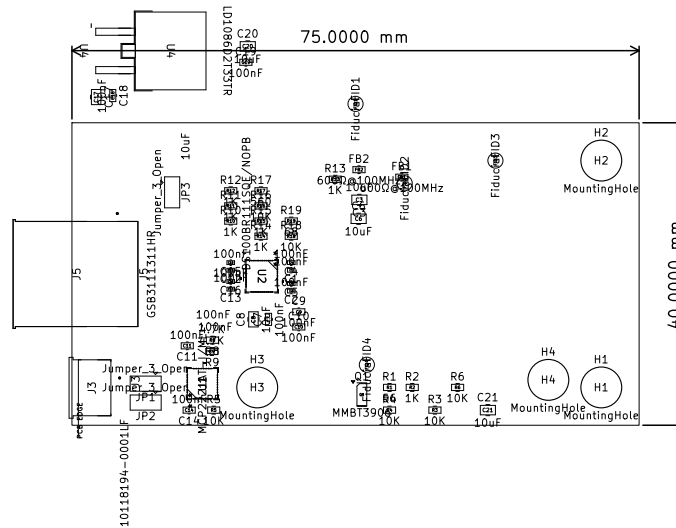
Track width: 0.10mm (4 mil)
Track spacing: 0.10mm (4 mil)
Via drill/dia: 0.30mm / 0.50mm
Annular ring: 0.10mm
Hole-to-hole: 0.25mm
Copper-to-edge: 0.50mm
Solder mask bridge: 0.10mm
Mask-to-trace: 0.09mm

VIA PLUGGING

Fill vias with soldermask (opaque)
Max via diameter for plugging: 0.5mm

NOTES

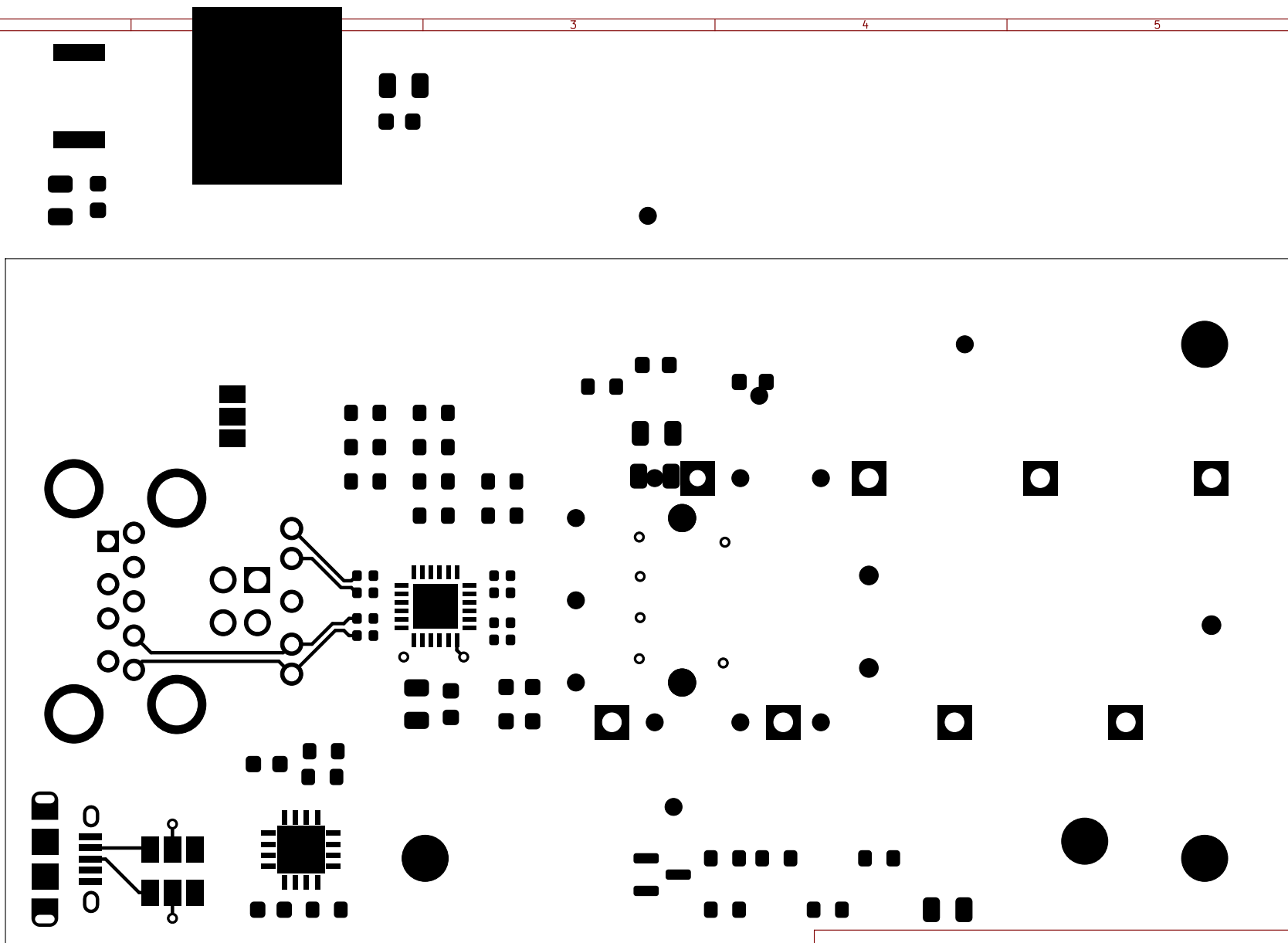
1. All dimensions in mm unless noted
2. Panelize per fab requirements
3. No blind/buried vias required



USB 3.0 to SFP+ Board LAYER: Overview

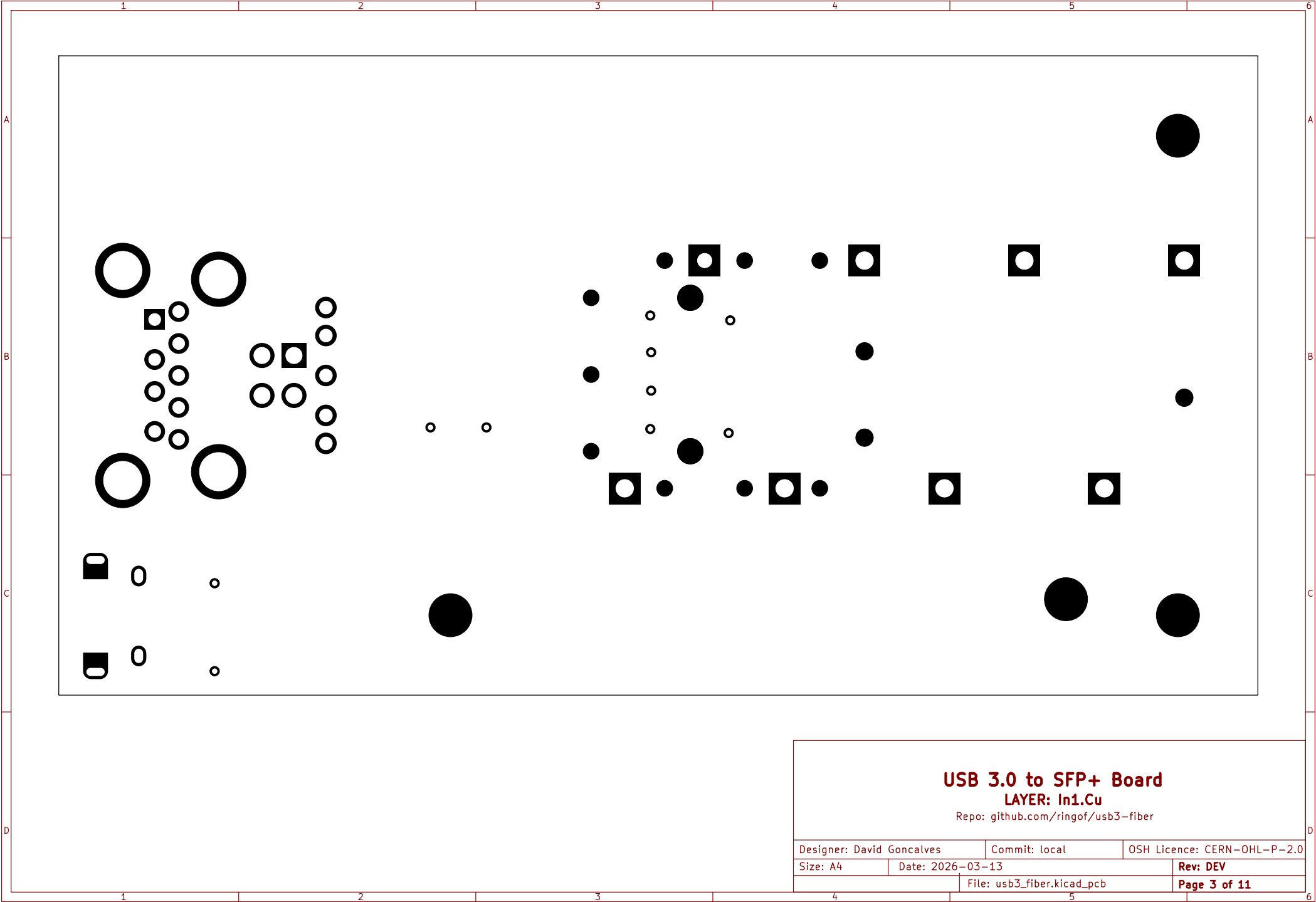
Repo: github.com/ringof/usb3-fiber

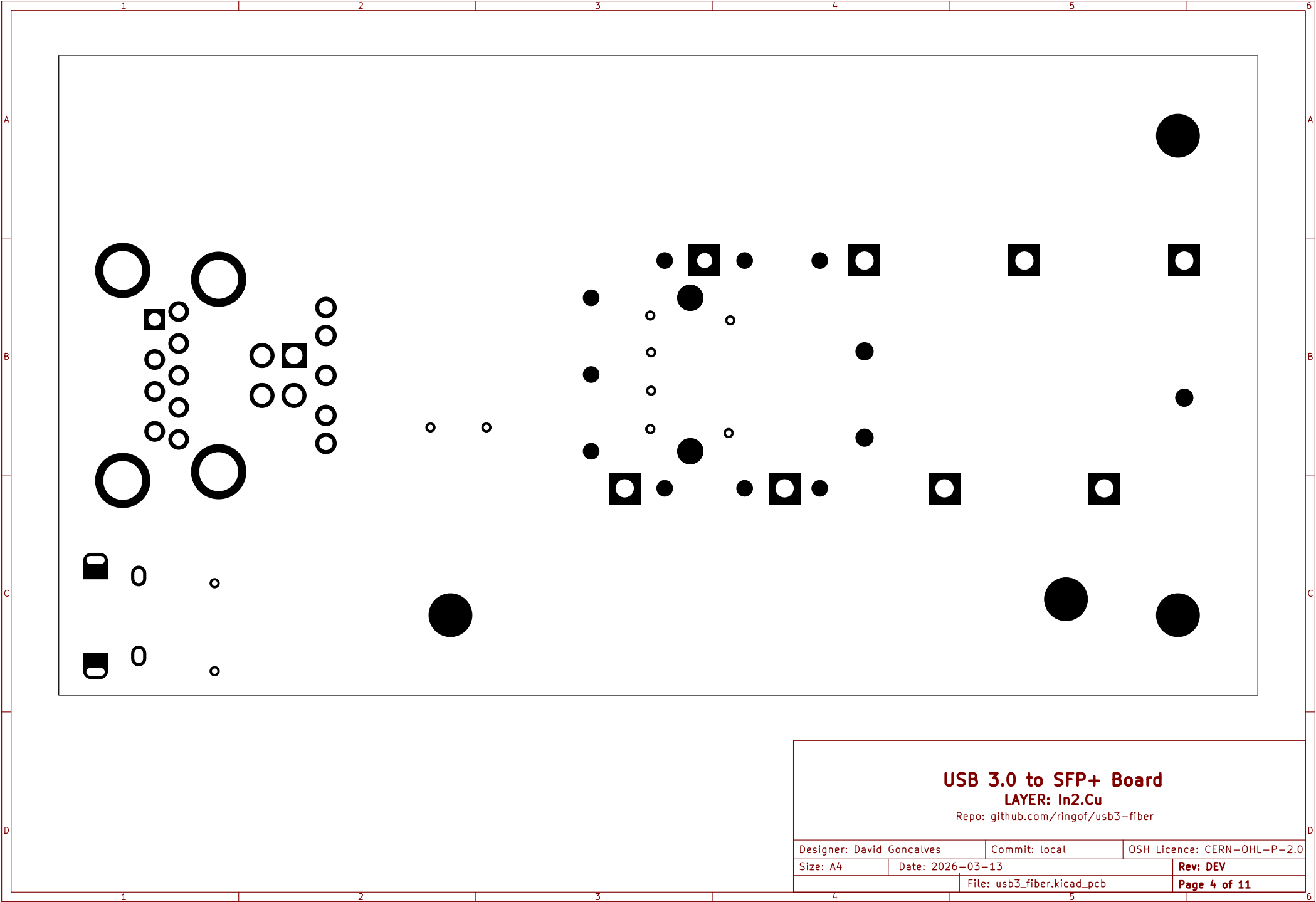
Designer: David Goncalves	Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13	Rev: DEV
	File: usb3_fiber.kicad_pcb	Page 1 of 11



USB 3.0 to SFP+ Board
LAYER: F.Cu
Repo: github.com/ringof/usb3-fiber

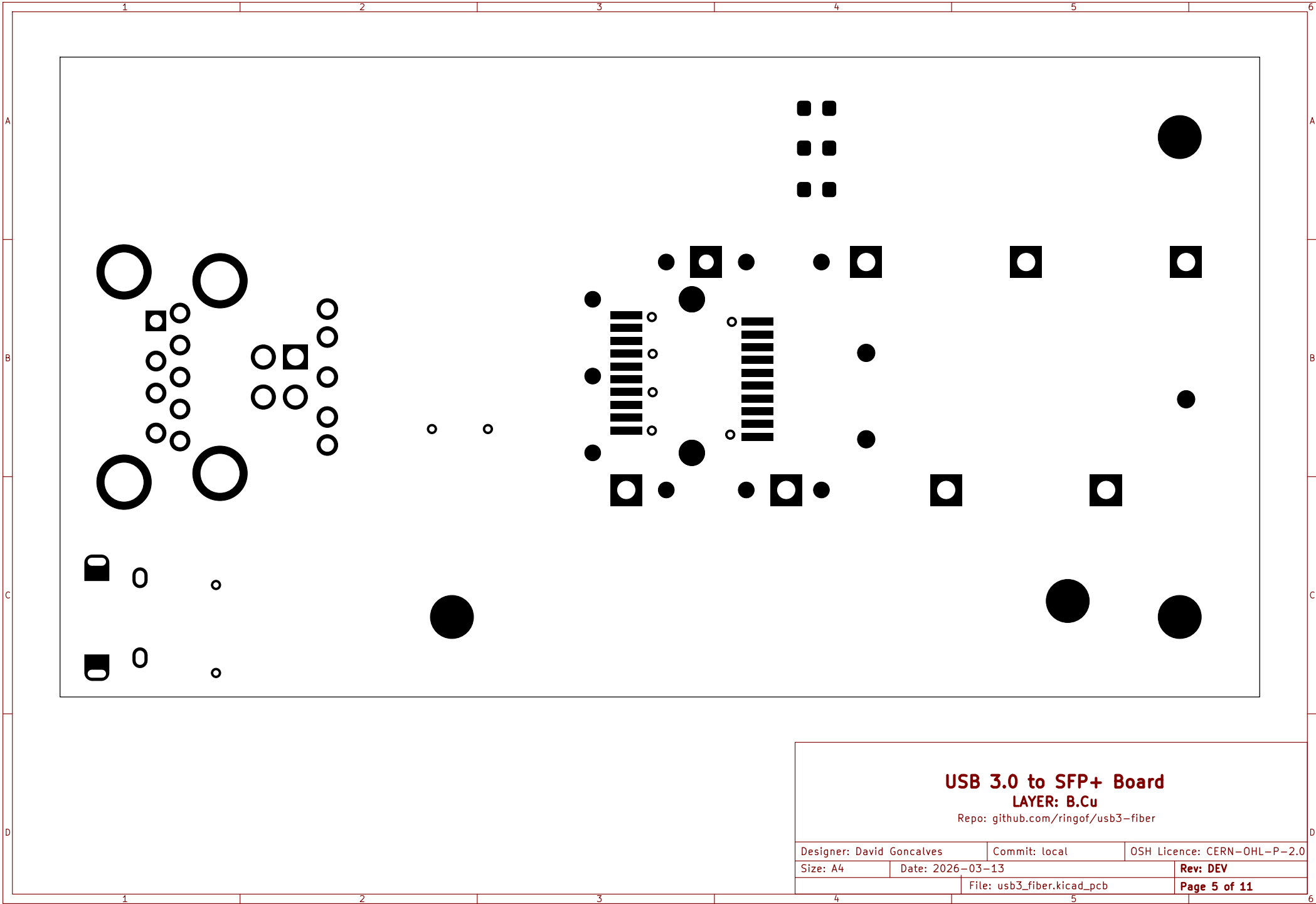
Designer: David Goncalves		Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13		Rev: DEV
		File: usb3_fiber.kicad_pcb	Page 2 of 11





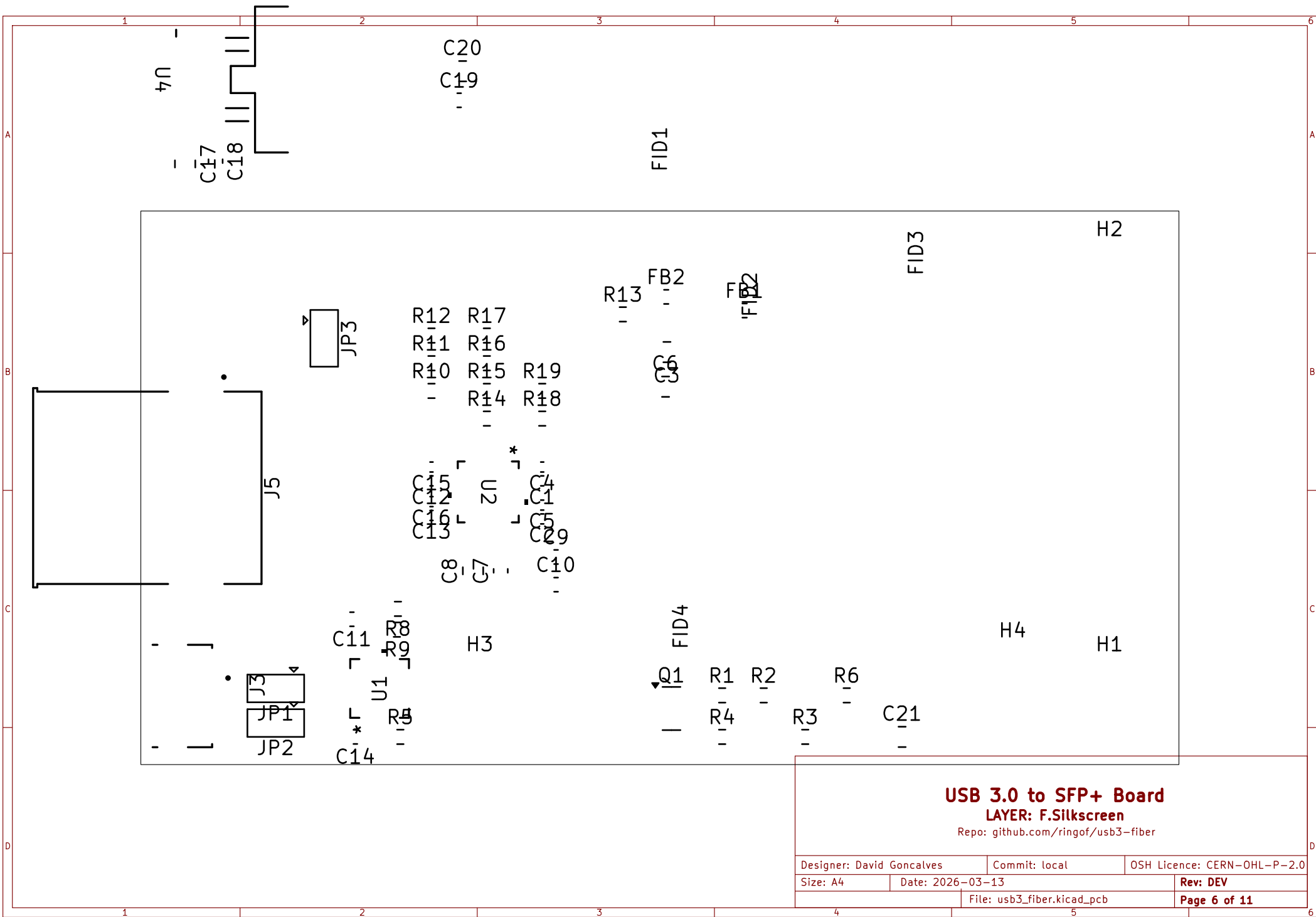
USB 3.0 to SFP+ Board
LAYER: In2.Cu
Repo: github.com/ringof/usb3-fiber

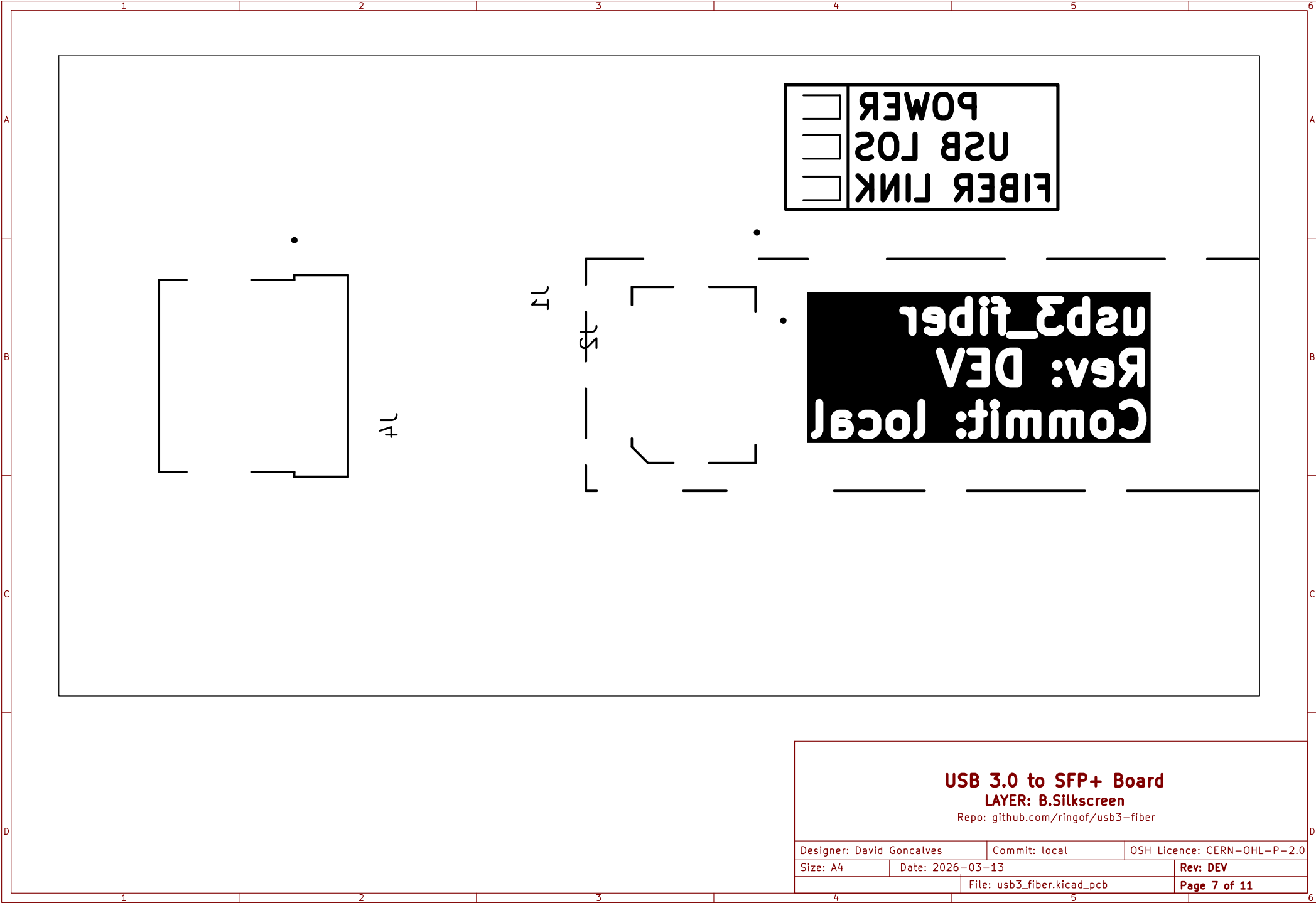
Designer: David Goncalves		Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13		Rev: DEV
		File: usb3_fiber.kicad_pcb	Page 4 of 11



USB 3.0 to SFP+ Board
LAYER: B.Cu
Repo: github.com/ringof/usb3-fiber

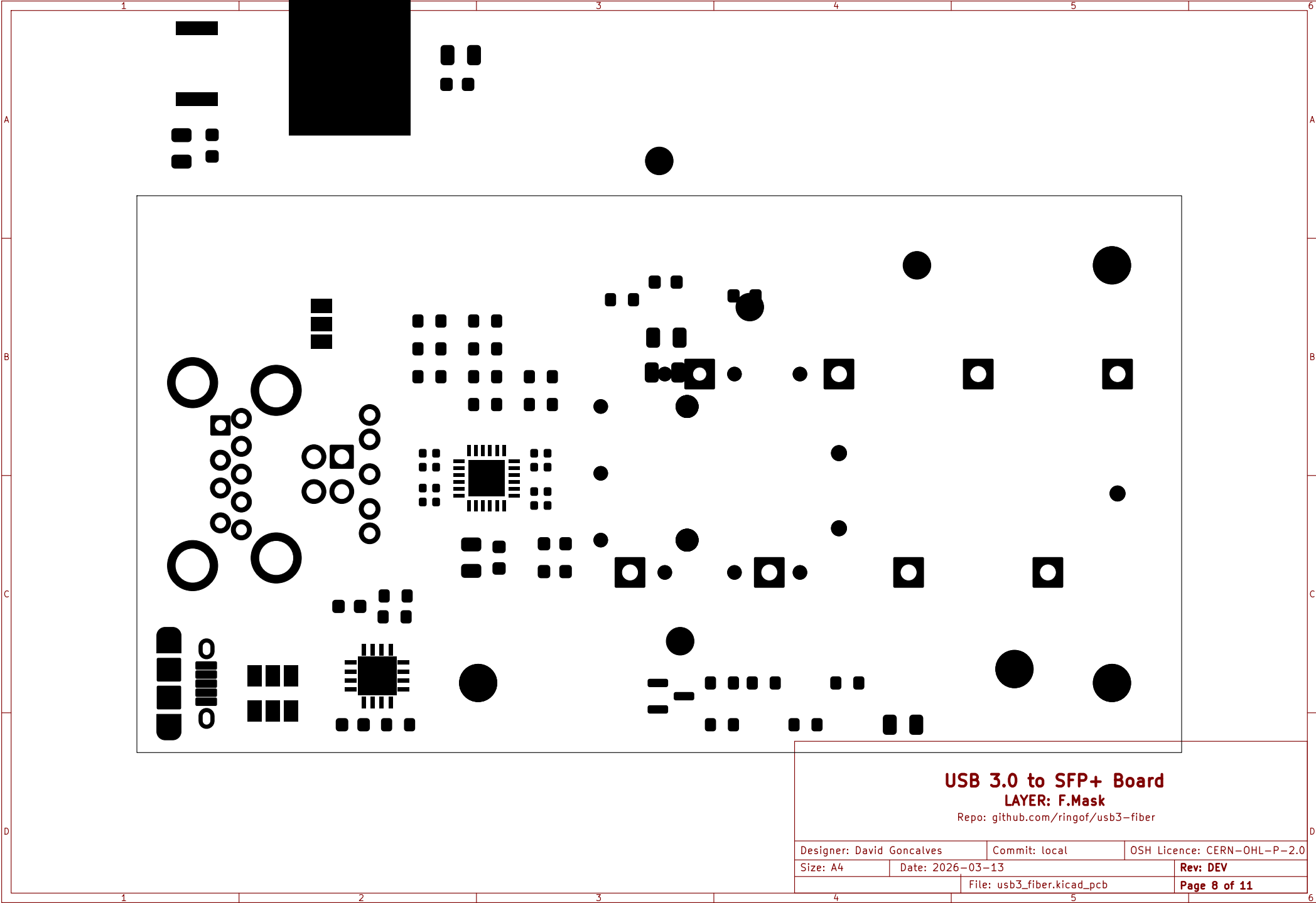
Designer: David Goncalves		Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13		Rev: DEV
		File: usb3_fiber.kicad_pcb	Page 5 of 11





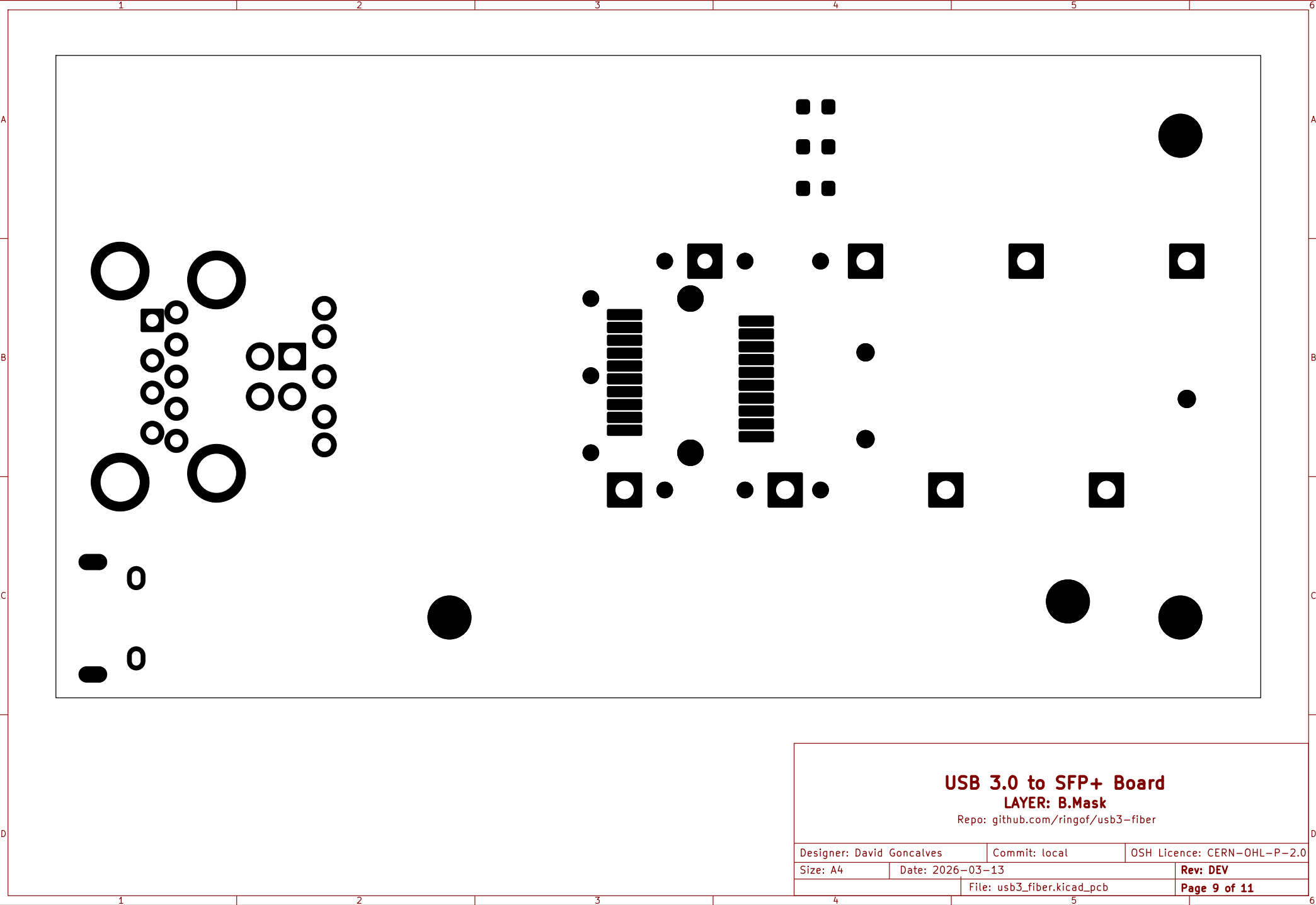
USB 3.0 to SFP+ Board
LAYER: B.Silkscreen
Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves		Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13		Rev: DEV
		File: usb3_fiber.kicad_pcb	Page 7 of 11



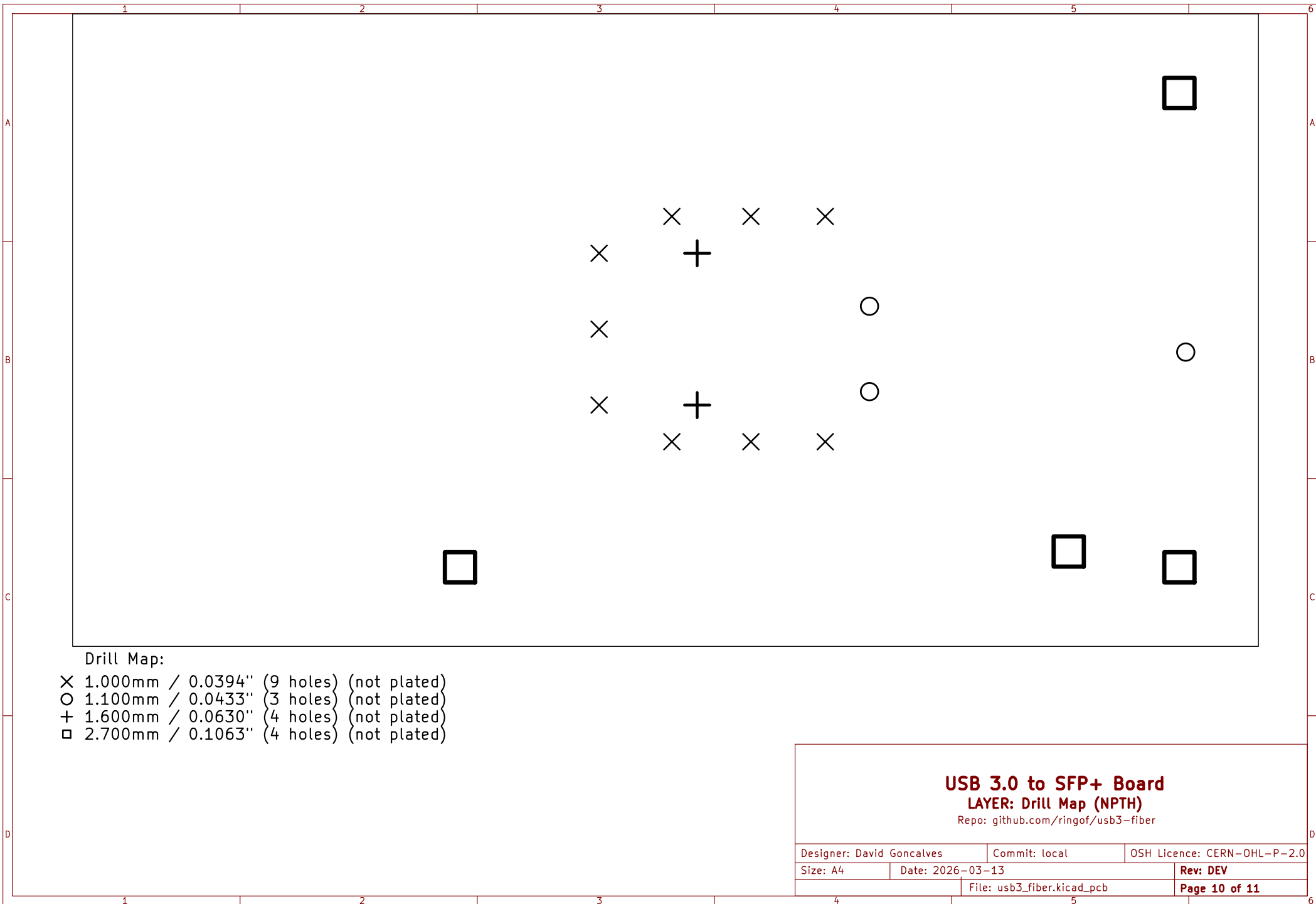
Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves		Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13		Rev: DEV
	File: usb3_fiber.kicad_pcb		Page 8 of 11



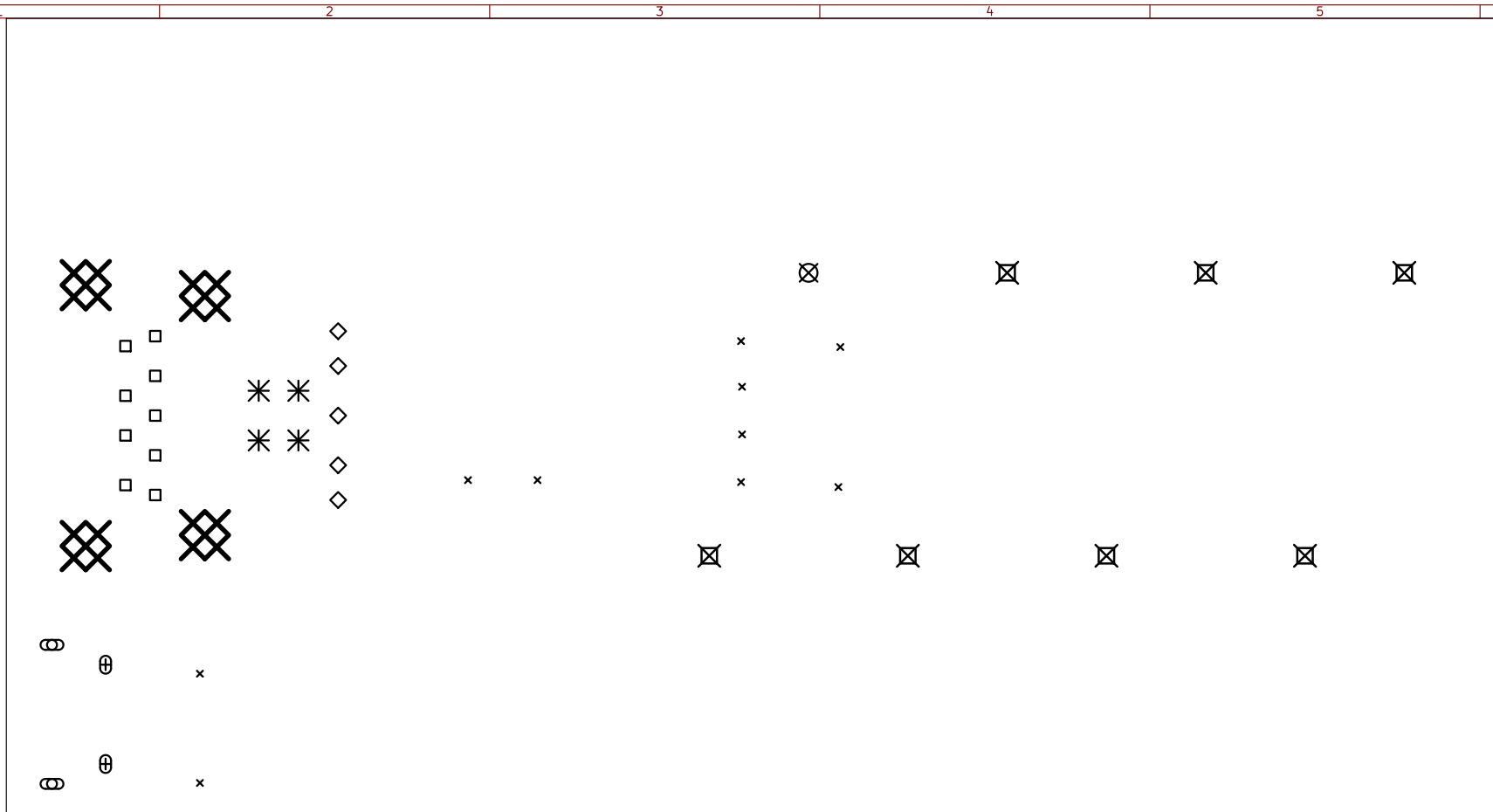
USB 3.0 to SFP+ Board
LAYER: B.Mask
Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves		Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13		Rev: DEV
		File: usb3_fiber.kicad_pcb	Page 9 of 11



USB 3.0 to SFP+ Board
LAYER: Drill Map (NPTH)
Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves		Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13		Rev: DEV
		File: usb3_fiber.kicad_pcb	Page 10 of 11



Drill Map:

x	0.300mm	/	0.0118"	(10 holes)
o	0.500mm	/	0.0197"	(0 holes + 2 slots)
+	0.550mm	/	0.0217"	(0 holes + 2 slots)
□	0.750mm	/	0.0295"	(9 holes)
◇	0.800mm	/	0.0315"	(5 holes)
⊗	0.900mm	/	0.0354"	(1 hole)
*	1.020mm	/	0.0402"	(4 holes)
⊠	1.100mm	/	0.0433"	(7 holes)
⊞	2.400mm	/	0.0945"	(4 holes)

USB 3.0 to SFP+ Board

LAYER: Drill Map (PTH)

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves		Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13		Rev: DEV
		File: usb3_fiber.kicad_pcb	Page 11 of 11